MATH 152 — PART 1

Number Systems, Inequalities, Absolute Values, and Point Sets
Detailed Notes with Tips and Examples

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1.0 Introduction to Number Systems

The Hierarchy of Numbers

Understanding the relationship between different number sets is fundamental to mathematics. Think of it as a Russian nesting doll — each set contains the previous ones.

```
R (Real Numbers)
┌──────────────────┐
│   Q (Rational)   │
│  ┌────────────┐  │
│  │  Z (Integers)│ │
│  │ ┌──────────┐ │ │
│  │ │ W (Whole)│ │ │
│  │ │┌───────┐│ │ │
│  │ ││N (Nat)││ │ │
│  │ │└───────┘│ │ │
│  │ └──────────┘ │ │
│  └────────────┘  │
└──────────────────┘
```

Detailed Breakdown

Natural Numbers (N) = {1, 2, 3, 4, ...}

Key Properties:

- Also called "counting numbers"
- **Closed** under addition: $2 + 3 = 5 \in N$
- **Closed** under multiplication: $2 \times 3 = 6 \in N$
- **NOT closed** under subtraction: $2 - 3 = -1 \notin N$
- **NOT closed** under division: $2 \div 3 = 2/3 \notin N$

💡**TIP:** "Closed" means performing the operation on any two numbers in the set always gives a result that's ALSO in the set.

Whole Numbers (W) = {0, 1, 2, 3, 4, ...}

Key Properties:

- $N \cup \{0\} = W$
- Adding zero as a "placeholder"

Integers (Z) = {..., −3, −2, −1, 0, 1, 2, 3, ...}

Key Properties:

- **Closed** under addition: $(-3) + 5 = 2 \in Z$
- **Closed** under multiplication: $(-2) \times 4 = -8 \in Z$
- **Closed** under subtraction: $3 - 5 = -2 \in Z$
- **NOT closed** under division: $3 \div 2 = 1.5 \notin Z$

Subsets of Z:

- **Positive integers**: $Z^+ = \{1, 2, 3, \ldots\}$
- **Negative integers**: $Z^- = \{\ldots, -3, -2, -1\}$
- **Non-negative integers**: $\{0\} \cup Z^+ = \{0, 1, 2, 3, \ldots\}$

💡**TIP:** Remember: $Z = Z^- \cup \{0\} \cup Z^+$

Rational Numbers (Q) = {a/b : a,b ∈ Z, b ≠ 0}

Key Properties:

- **Closed** under ALL arithmetic operations (+, −, ×, ÷)
- **Exception:** Division by zero is undefined!

Important Notes:

1. $0 \in Q$ (since $0 = 0/1$, $b \neq 0$)
2. Every integer is rational ($a = a/1$)
3. So $Z \subset Q$

Examples of Rational Numbers: 3/4 (proper fraction), 5/2 (improper fraction), 0.75 (terminating decimal), 0.333... (recurring decimal)

> 🔑**KEY INSIGHT:** A number is rational if it can be expressed as a fraction of two integers (with denominator $\neq 0$).

Irrational Numbers (I)

Definition: Numbers that CANNOT be expressed in the form a/b where $a,b \in Z$.

Common Examples: $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$ (square roots of non-perfect squares); $\pi \approx 3.14159...$; $e \approx 2.71828...$; $\sqrt{2} + 1$ (sum of rational and irrational is irrational)

Decimal Representation:

Type	Decimal Form	Example
Terminating decimal	Ends after finite digits	1/2 = 0.5
Recurring decimal	Repeats pattern	1/3 = 0.333...
Non-terminating, non-recurring	Never ends, no pattern	$\sqrt{2}$ = 1.41421356...

The Proof That √2 is Irrational

This is a CLASSIC proof by contradiction. Memorize the structure — it appears in exams frequently!

STEP-BY-STEP PROOF

```
Step 1: Assume the opposite
Assume √2 is rational.
Then √2 = a/b, where a,b ∈ Z, b ≠ 0, and a/b is in lowest terms.

Step 2: Square both sides
2 = a²/b²
⇒ a² = 2b²

Step 3: Analyze a²
a² = 2b² → a² is even (multiple of 2)
If a² is even, then a is even (only even numbers square to even numbers)
So a = 2c for some integer c.

Step 4: Substitute
(2c)² = 2b²
4c² = 2b²
b² = 2c²

Step 5: Analyze b²
b² = 2c² → b² is even → b is even

Step 6: Contradiction!
Both a and b are even.
So a/b can be simplified (divided by 2).
But we assumed a/b was in lowest terms.

CONTRADICTION!
Therefore, √2 is irrational.
```

> 💡**TIP:** This proof uses the following facts: (1) A number is even → its square is even. (2) A square is even → the number is even. (3) If both numerator and denominator are even, the fraction can be simplified.

1.1 Operations with Real Numbers

The Field Axioms

A set that satisfies these rules is called a **FIELD**. The real numbers R form a field.

Rule	Name	Mathematical Statement
1	Closure (+)	$a + b \in R$
2	Closure (×)	$ab \in R$
3	Commutative (+)	$a + b = b + a$
4	Commutative (×)	$ab = ba$
5	Associative (+)	$(a+b)+c = a+(b+c)$
6	Associative (×)	$(ab)c = a(bc)$
7	Distributive	$a(b+c) = ab + ac$
8	Identity (+)	$a + 0 = 0 + a = a$
9	Identity (×)	$a\cdot 1 = 1\cdot a = a$
10	Inverse (+)	$a + (-a) = 0$
11	Inverse (×)	$a\cdot a^{-1} = 1$ $(a \neq 0)$

💡**TIP:** These axioms are the "rules of the game" for algebra. Every algebraic manipulation relies on these fundamental laws.

1.2 Inequalities

Fundamental Theorems

Theorem 1: Law of Trichotomy

For any $a, b \in R$, exactly ONE of the following is true: $a > b$ OR $a = b$ OR $a < b$

💡**TIP:** This simply means any two numbers can be compared — one is greater, equal, or less than the other.

Theorem 2: Law of Transitivity

If $a > b$ and $b > c$, then $a > c$
Example: If $5 > 3$ and $3 > 1$, then $5 > 1$

💡**TIP:** This allows us to "chain" inequalities together.

Theorem 3: Addition Property

If $a > b$, then $a + c > b + c$
Example: If $5 > 3$, then $5 + 2 > 3 + 2$ $(7 > 5)$

Theorem 4: Multiplication by Positive Number

If $a > b$ and $c > 0$, then $ac > bc$
Example: If $5 > 3$, then $5 \times 2 > 3 \times 2$ $(10 > 6)$

Theorem 5: Multiplication by Negative Number

If $a > b$ and $c < 0$, then $ac < bc$

⚠**CRITICAL WARNING:** When multiplying or dividing an inequality by a **negative** number, you MUST flip the inequality sign!

```
Example:
-2x < 6
Divide both sides by -2 (negative!)
x > -3   ← SIGN FLIPPED!
```

💡**TIP:** Many students forget this. Always remember: NEGATIVE × SIGN FLIP!

Solving Quadratic Inequalities

Method 1: Case Analysis (Most Common)

EXAMPLE — Solve $x^2 - x - 2 > 0$

```
Step 1: Factor
(x + 1)(x - 2) > 0

Step 2: For product > 0, both factors must have same sign
Case 1: Both positive
x + 1 > 0 AND x - 2 > 0
x > -1 AND x > 2
x > 2 (intersection)

Case 2: Both negative
x + 1 < 0 AND x - 2 < 0
x < -1 AND x < 2
x < -1 (intersection)

Step 3: Combine
{x : x < -1} ∪ {x : x > 2}
```

💡**TIP:** For > 0 (positive product), factors must have SAME sign. For < 0 (negative product), factors must have OPPOSITE signs.

Method 2: Graphical Method

Steps: (1) Factor the expression. (2) Find roots (where expression = 0). (3) Sketch the parabola. (4) Identify regions where $y > 0$ or $y < 0$.

EXAMPLE — Solve $x^2 - x - 2 > 0$

```
f(x) = (x + 1)(x - 2)
Roots: x = -1, x = 2

The parabola opens upward (coefficient of x² is positive)
Regions:
- x < -1: f(x) > 0 ✓
- -1 < x < 2: f(x) < 0 ✗
- x > 2: f(x) > 0 ✓

Solution: x < -1 or x > 2
```

```
     ▲ f(x)
     │
─────┼─────┬──────────────► x
     │    / \
     │   /   \
─────┼──/─────\───
-1   │ /  2    \
     │/         \
     │/          \
f(x)>0 at x<-1  f(x)>0 at x>2
```

Method 3: Sign Table Method

Steps: (1) Find roots. (2) Create a table with intervals between roots. (3) Determine sign of each factor in each interval. (4) Determine overall sign.

EXAMPLE — Solve $x^2 - x - 2 > 0$

$f(x) = (x + 1)(x - 2)$, Roots: $x = -1$, $x = 2$

Interval	$x < -1$	$-1 < x < 2$	$x > 2$
$x + 1$	−	+	+
$x - 2$	−	−	+
f(x)	**+**	**−**	**+**

Solution: $x < -1$ or $x > 2$

💡**TIP:** The sign table method is the most systematic and least prone to errors!

Special Cases

Case: Product ≥ 0 or ≤ 0

Example: $x^2 - x - 2 \geq 0$ — **Solution includes roots!**
$\{x : x \leq -1\} \cup \{x : x \geq 2\}$

💡**TIP:** ≥ includes the roots, > excludes them.

Case: Negative Leading Coefficient

Example: $9 - 3x - 2x^2 \leq 0$

Option 1 — Multiply by −1:

```
9 - 3x - 2x² ≤ 0
Multiply by -1 (FLIP sign!)
2x² + 3x - 9 ≥ 0
```

Option 2 — Factor directly:

```
9 - 3x - 2x² = (3 - 2x)(3 + x) ≤ 0
```

💡**TIP:** For parabolas: Coefficient of $x^2 > 0$ → "smile" shape ∪ (opens upward). Coefficient of $x^2 < 0$ → "frown" shape ∩ (opens downward).

Common Mistakes to Avoid

Mistake	Correct Way
Forgetting to flip sign when multiplying by negative	Always flip when multiplying/dividing by negative
Using AND instead of OR	Different regions use OR; overlapping regions use AND
Including roots for strict inequalities	$>$ and $<$ exclude roots; $\geq$ and $\leq$ include roots
Not checking the sign of a	Always consider that a could be negative in $ax^2 + bx + c$

1.3 Absolute Value

Definition

The absolute value of x, denoted |x|, is the **distance** of x from zero on the number line.

```
|x| = { x,   if x ≥ 0
      { -x,  if x < 0
```

🔑**KEY INSIGHT:** Absolute value is NEVER negative. It only asks "how far?" not "in which direction?"

Examples: $|3| = 3$ (3 units to the right of 0). $|-3| = 3$ (3 units to the left of 0). $|0| = 0$ (0 units from 0).

Properties of Absolute Value

Property	Formula	Explanation
Non-negativity	$\lvert x\rvert \geq 0$	Absolute value is never negative
Product	$\lvert xy\rvert = \lvert x\rvert\lvert y\rvert$	Absolute value distributes over multiplication
Triangle Inequality	$\lvert x + y\rvert \leq \lvert x\rvert + \lvert y\rvert$	Sum of absolutes is at least absolute of sum
Reverse Triangle	$\lvert x - y\rvert \geq \lvert\lvert x\rvert - \lvert y\rvert\rvert$	Difference inequality
Zero	$\lvert x\rvert = 0 \Leftrightarrow x = 0$	Only zero has absolute value zero

Proving the Properties

PROOF OF |xy| = |x||y|

```
Case 1: x > 0, y > 0
xy > 0 → |xy| = xy = |x||y|

Case 2: x < 0, y > 0
xy < 0 → |xy| = -(xy) = (-x)y = |x||y|

Case 3: x < 0, y < 0
xy > 0 → |xy| = xy = (-x)(-y) = |x||y|

Case 4: x = 0 or y = 0
|xy| = 0 = |x||y|

Therefore, |xy| = |x||y| for all real x,y.
```

PROOF OF TRIANGLE INEQUALITY: |x + y| ≤ |x| + |y|

```
Case 1: x ≥ 0, y ≥ 0
x + y ≥ 0 → |x + y| = x + y = |x| + |y|

Case 2: x < 0, y < 0
x + y < 0 → |x + y| = -(x + y) = (-x) + (-y) = |x| + |y|

Case 3: x ≥ 0, y < 0 (and |x| > |y|)
x + y > 0 → |x + y| = x + y = |x| - |y| < |x| + |y|

Case 4: x ≥ 0, y < 0 (and |x| < |y|)
x + y < 0 → |x + y| = -(x + y) = |y| - |x| < |x| + |y|

Therefore, |x + y| ≤ |x| + |y| for all real x,y.
```

Geometric Interpretation

```
Number Line:
    ←---|----|----|----|----|----|----→
       -3   -2   -1    0    1    2    3
       └──────────┘
       |x| = 3 for x = -3 or x = 3
```

Distance Between Points:

```
|x - y| = distance between x and y on the number line
|x - y| = |y - x| (symmetric)
```

Example: $|5 - 2| = 3$ (distance from 2 to 5). $|2 - 5| = 3$ (distance from 5 to 2).

Extended Properties

Product of n numbers: $|x_1 x_2 \ldots x_n| = |x_1||x_2|\ldots|x_n|$

Sum of n numbers: $|x_1 + x_2 + \ldots + x_n| \leq |x_1| + |x_2| + \ldots + |x_n|$

Simplifying Absolute Value Expressions

Expression	Simplified	Explanation
$\lvert 0 - 6 \rvert$	6	$\lvert -6 \rvert = 6$
$\lvert 5 - 2 \rvert$	3	$\lvert 3 \rvert = 3$
$\lvert 2 - 5 \rvert$	3	$\lvert -3 \rvert = 3$
$\lvert 0(-4) \rvert$	0	$\lvert 0 \rvert = 0$
$\lvert 2 + 3(-4) \rvert$	10	$\lvert 2 + (-12) \rvert = \lvert -10 \rvert = 10$
$-\lvert -4 \rvert$	-4	$-(4) = -4$
$-\lvert (-3)^2 \rvert$	-9	$-\lvert 9 \rvert = -9$

💡**TIP:** Always evaluate inside the absolute value bars first!

1.4 Point Sets

Definition

A **point set** is any set whose members are real numbers. All elements can be located on the real number line.

Examples: $\{0, 1, 2, 3, 4, 5\}$ (finite point set); $\{x : -1 < x < 0\}$ (infinite point set — interval)

Intervals

Bounded Intervals

Type	Notation	Set Notation	Example
Closed	$[a,b]$	$\{x : a \le x \le b\}$	$[0,1]$ includes 0 and 1
Open	(a,b)	$\{x : a < x < b\}$	$(0,1)$ excludes 0 and 1
Half-open	$(a,b]$	$\{x : a < x \le b\}$	$(0,1]$ includes 1 only
Half-open	$[a,b)$	$\{x : a \le x < b\}$	$[0,1)$ includes 0 only

```
Number Line Representation:
[a,b]     ═══════●═══════●═══════   (filled circles = included)
(a,b)     ───────○═══════○───────   (empty circles = excluded)
(a,b]     ───────○═══════●═══════
[a,b)     ═══════●═══════○───────
```

Unbounded Intervals

Type	Notation	Set Notation
Open left	$(-\infty, b)$	$\{x : x < b\}$
Closed left	$(-\infty, b]$	$\{x : x \le b\}$
Open right	(a, ∞)	$\{x : x > a\}$
Closed right	$[a, \infty)$	$\{x : x \ge a\}$
All real	$(-\infty, \infty)$	R

Important: ∞ and $-\infty$ are NOT real numbers — they're just notation indicating "unbounded."

💡**TIP:** ∞ always gets parentheses (never a bracket) because it's not a number that can be included.

Neighbourhoods

δ-Neighbourhood of a point a

```
Nδ(a) = {x : |x - a| < δ} = (a - δ, a + δ)
```

```
Geometric Interpretation:
    ←───(a-δ)────[a]────(a+δ)───→
    └─────────────2δ──────────────┘
    Center = a, Length = 2δ
```

Example: δ-neighbourhood of 2 with $\delta = 0.5$: $N_{0.5}(2) = (1.5, 2.5)$

Deleted δ-Neighbourhood

```
Nδ*(a) = {x : 0 < |x - a| < δ} = (a - δ, a) ∪ (a, a + δ)
```

Difference: Excludes the point a itself.

```
    ←───(a-δ)────(a)────(a+δ)───→
                    ×
            (point a is removed)
```

Limit Points (Cluster Points)

Definition

A number l is a **limit point** of set A if EVERY deleted δ-neighbourhood of l contains elements of A.

In other words: For every $\delta > 0$, $\exists\, x \in A$ such that $0 < |x - l| < \delta$.

```
    ────(l-δ)────(l)────(l+δ)────
                  ×
         [Element of A] [Element of A]
```

Key Characteristics:

1. δ can be as small as we please
2. Every δ-neighbourhood contains infinitely many elements of A
3. Limit points don't have to be in the set itself

Examples of Limit Points

Example 1: $A = (0, 1)$

Limit points: 0 and 1 (and all points in between)

Why is 0 a limit point? Every δ-neighbourhood of 0, i.e. $(-\delta, \delta)$, will contain elements of $(0,1)$ for any $\delta > 0$. But 0 itself is NOT in the set (it's open).

Example 2: Finite Set $\{1, 2, 3, 4, 5\}$

Limit points: NONE

Why? For any suggested number l, we can choose δ small enough that the deleted δ-neighbourhood contains no elements of the set.

Example 3: $N = \{1, 2, 3, 4, ...\}$

Limit points: NONE

Why? The natural numbers "run away" to infinity. For any proposed limit, we can find a small enough neighbourhood with no natural numbers.

Example 4: $A = \{1, 1/2, 1/3, 1/4, ...\}$

Limit point: 0

Why? Elements get arbitrarily close to 0 but never reach it.

Example 5: $A = Q$ (Rational Numbers)

Limit points: ALL real numbers

Why? Every interval contains rational numbers.

Closed Sets

Definition: A set is **closed** if it contains ALL its limit points.

Set	Closed?	Why?
$[0,1]$	Yes	Contains 0 and 1
$(0,1)$	No	Does NOT contain 0 and 1
$\{1, 1/2, 1/3, ...\} \cup \{0\}$	Yes	Contains limit point 0
$\{1, 1/2, 1/3, ...\}$	No	Does NOT contain limit point 0

Bounds

Definitions

Term	Definition	Example for $A = [0,1]$
Bounded above	$\exists$ M such that $x \leq M, \forall x \in A$	$M = 2, 3, 5$, etc.
Upper bound	Any M that bounds A above	2, 3, 5, ...
Least upper bound (l.u.b.)	Smallest upper bound	1
Bounded below	$\exists$ m such that $x \geq m, \forall x \in A$	$m = -1, -2, -5$, etc.
Lower bound	Any m that bounds A below	$-1, -2, -5, ...$
Greatest lower bound (g.l.b.)	Largest lower bound	0

```
Visual Representation:
    m (lower bound)    M (upper bound)

    A = [0,1]
    g.l.b. = 0    l.u.b. = 1
```

Examples of Bounds

Set	Upper Bounds	l.u.b.	Lower Bounds	g.l.b.	Bounded?
$[0,1]$	1, 2, 3, ...	1	$0, -1, -2, ...$	0	Yes
$(0,1)$	1, 2, 3, ...	1	$0, -1, -2, ...$	0	Yes*
$\{1, 1/2, 1/3, ...\}$	1, 2, 3, ...	1	$0, -1, -2, ...$	0	Yes
$N = \{1, 2, 3, ...\}$	NONE	Does not exist	$1, 0, -1, ...$	1	No (not bounded above)

💡**TIP:** *For $(0,1)$: l.u.b. and g.l.b. may NOT be in the set!

Bolzano-Weierstrass Theorem

🔑**KEY INSIGHT:** Every bounded infinite set has at least one limit point.

What This Means: If a set is bounded and infinite, it must "accumulate" somewhere. This is a fundamental theorem in real analysis.

Set	Bounded?	Infinite?	Limit Point?
A = {1, 1/2, 1/3, 1/4, ...}	Yes ($0 \le x \le 1$)	Yes	0 ✓ (satisfies theorem)
A = {1, 2, 3, 4, ...}	No (not bounded above)	Yes	None ✓ (theorem doesn't apply)
A = (0,1)	Yes	Yes	[0,1] (infinitely many) ✓

Practice Problems and Solutions

PROBLEM 1

Find the truth set of $x^2 - 7x + 10 \geq 0$

Solution:

```
x² - 7x + 10 = (x - 2)(x - 5)
Roots: x = 2, x = 5

Interval      x < 2     2 < x < 5     x > 5
─────────────────────────────────────────────
x - 2           -           +            +
x - 5           -           -            +
─────────────────────────────────────────────
f(x)            +           -            +

Since we want f(x) ≥ 0:
{x : x ≤ 2} ∪ {x : x ≥ 5}
```

PROBLEM 2

Solve $6x^2 - x - 35 \leq 0$

Solution:

```
6x² - x - 35 = (3x + 7)(2x - 5)
Roots: x = -7/3, x = 5/2

Interval      x < -7/3   -7/3 < x < 5/2   x > 5/2
─────────────────────────────────────────────────────
3x + 7           -              +              +
2x - 5           -              -              +
─────────────────────────────────────────────────────
f(x)             +              -              +

Since we want f(x) ≤ 0:
{x : -7/3 ≤ x ≤ 5/2}
```

PROBLEM 3

Solve $(1 - 2x)(x + 2) < 0$

Solution:

```
Roots: 1 - 2x = 0 → x = 1/2
        x + 2 = 0 → x = -2

Interval      x < -2      -2 < x < 1/2     x > 1/2
─────────────────────────────────────────────────────
1 - 2x          +              +              -
x + 2           -              +              +
─────────────────────────────────────────────────────
f(x)            -              +              -

Since we want f(x) < 0:
{x : x < -2} ∪ {x : x > 1/2}
```

PROBLEM 4

Find the truth set of $35 - 2x - x^2 \geq 0$

Solution (Method 1 — Multiply by −1):

```
35 - 2x - x² ≥ 0
Multiply by -1 (FLIP sign):
x² + 2x - 35 ≤ 0
(x + 7)(x - 5) ≤ 0

Roots: x = -7, x = 5
For ≤ 0: -7 ≤ x ≤ 5
```

Solution (Method 2 — Work directly):

```
35 - 2x - x² = -(x² + 2x - 35) = -(x + 7)(x - 5)
For ≥ 0: need (x + 7)(x - 5) ≤ 0
-7 ≤ x ≤ 5
```

PROBLEM 5

Find the truth set of $3x^2 - x + 2 > 0$

Solution:

```
Discriminant: Δ = b² - 4ac = (-1)² - 4(3)(2) = 1 - 24 = -23
Since Δ < 0, the parabola never crosses x-axis.
a = 3 > 0 → parabola opens upward → always positive.

Truth set: All real numbers R
```

TIP: If discriminant is negative and $a > 0$, the quadratic is always positive!

PROBLEM 6

Find the truth set of $(3 - 2x)(4 - 2x) \leq 0$

Solution:

```
Roots: 3 - 2x = 0 → x = 3/2
        4 - 2x = 0 → x = 2

Interval      x < 3/2     3/2 < x < 2     x > 2
────────────────────────────────────────────────
3 - 2x          +              -              -
4 - 2x          +              +              -
────────────────────────────────────────────────
f(x)            +              -              +

Since we want f(x) ≤ 0:
{x : 3/2 ≤ x ≤ 2}
```

PROBLEM 7

Find the cluster point(s) of A = {1, 1/2, 1, 1/3, 1, 1/4, 1, 1/5, ...}

Solution:

```
The sequence is: 1, 1/2, 1, 1/3, 1, 1/4, 1, 1/5, ...
Elements: All numbers of the form 1/n (n ≥ 2) and the number 1.

Limit points:
- 1 is a limit point (repeated infinitely)
- 0 is a limit point (1/n → 0)

So the cluster points are: {0, 1}
```

PROBLEM 8

Does T = {1, 2, 3, 4, 5} have a cluster point?

Solution:

```
No! T is a finite set.
Finite sets have NO cluster points.

For any proposed number l, we can choose δ small enough that
the deleted δ-neighbourhood contains no elements of T.
```

PROBLEM 9

Find the limit point(s) of $a_n = 1 + (-1)^n + 1/n$

Solution:

```
aₙ = 1 + (-1)ⁿ + 1/n

When n is even: (-1)ⁿ = 1
aₙ = 1 + 1 + 1/n = 2 + 1/n → 2

When n is odd: (-1)ⁿ = -1
aₙ = 1 - 1 + 1/n = 1/n → 0

So the sequence has two subsequences converging to different limits.

Limit points: {0, 2}
```

PROBLEM 10

Find the limit point(s) of A = {1/2, 1/3, 2/3, 1/4, 2/4, 3/4, 1/5, 2/5, 3/5, 4/5, ...}

Solution:

```
A = all fractions between 0 and 1 of the form k/n where 1 ≤ k < n.
This set is actually ALL rational numbers in (0,1).
Every rational number appears in this sequence.

Limit points of Q ∩ (0,1) = [0,1]

So the cluster points are all numbers in [0,1].
```

Common Mistakes and How to Avoid Them

Mistake 1: Forgetting to Flip the Inequality Sign

Wrong	Correct
$-3x < 6 \Rightarrow x < -2$ ✗	$-3x < 6 \Rightarrow x > -2$ ✓ (dividing by negative → flip sign!)

💡**TIP:** Always remember NEGATIVE = FLIP!

Mistake 2: Confusing AND with OR

Example: $(x + 1)(x - 2) > 0$

Wrong	Correct
$x > -1$ AND $x > 2 \rightarrow x > 2$ (only gives one part of the solution!)	$x < -1$ OR $x > 2$ (both positive and both negative cases)

💡**TIP:** For products > 0, use OR. For products < 0, use AND (between the roots).

Mistake 3: Including/Excluding Roots Incorrectly

For > or <: Exclude roots (open circles). **For ≥ or ≤:** Include roots (closed circles).

```
x² - 4 > 0 → x < -2 or x > 2
x² - 4 ≥ 0 → x ≤ -2 or x ≥ 2
```

Mistake 4: Not Checking if the Quadratic is Always Positive/Negative

⚠**CRITICAL WARNING:** Always check the discriminant! If $\Delta < 0$ and $a > 0$ → always positive. If $\Delta < 0$ and $a < 0$ → always negative.

Mistake 5: Confusing Cluster Points with Elements of the Set

A limit point doesn't have to be in the set!

Example: $(0,1)$ has limit points 0 and 1, even though $0, 1 \notin (0,1)$

Summary of Key Formulas

Number Systems

```
N ⊂ W ⊂ Z ⊂ Q ⊂ R
Q ∪ I = R
```

Inequalities

```
If a > b and c > 0 → ac > bc
If a > b and c < 0 → ac < bc
```

Absolute Value

```
|x| = { x, if x ≥ 0
      { -x, if x < 0

|xy| = |x||y|
|x + y| ≤ |x| + |y|
|x - y| ≥ ||x| - |y||
```

Intervals

```
[a,b] = {x : a ≤ x ≤ b}
(a,b) = {x : a < x < b}
(a,b] = {x : a < x ≤ b}
[a,b) = {x : a ≤ x < b}
```

Neighbourhoods

```
Nδ(a) = {x : |x - a| < δ}
Nδ*(a) = {x : 0 < |x - a| < δ}
```

— END OF PART 1 DETAILED NOTES —